THAKORN TANGPHADUNGRUTCH

thakorntang7@gmail.com - ledut122@gmail.com | Brentwood, Essex, United Kingdom

EDUCATION

Brentwood School, United Kingdom — Class of 2027
Standardized Tests: SAT: 1500 | AP Physics 1: 4 | AP Calculus BC: In-Progress

HONORS & AWARDS

- **First Place**, World Robot Olympiad (WRO), Vietnam National Round (2026)
  Competed in RoboSports (Double Tennis) at a national event with 350 teams and 1,500 participants; the team was selected as one of eight to represent Vietnam. Advanced to the International Round in Puerto Rico, USA.
- **Gold Medal**, Silicon Valley International Invention Festival - SVIIF (2025)
  Led an AI-based stock-market analysis project awarded Gold at an international festival in Santa Clara, California, USA, featuring over 200 inventions from 25 countries
- **Silver Award**, British Physics Olympiad Senior Competition (2025)
- **Gold Award**, Senior Maths Challenge (2024)
- **Year 11 Physics Award**, Brentwood School (2025)
- **Year 10 Academic Award**, Brentwood School (2024)

LEADERSHIP & COMMUNITY IMPACT

SafeFlow Vietnam (Project page) / ***Founder & President*** **Jan 2025 - Present**

A project integrating rescue technology, flood safety education, & relief supplies to support flood- and disaster-affected communities in Vietnam.

- Founded a technology-enabled flood- and water-safety initiative; led 2 field programs serving approximately 200 students and residents in Y Ty, Lao Cai, and Suoi Tuong floating village, Dong Nai, mobilizing more than $1,923 USD in aid and programming.
- Delivered water-safety education and introductory STEM activities; distributed safety equipment and school supplies and connected rescue-drone and smart-life-jacket concepts to needs identified in flood-prone communities.

Creativism Club (Project page) / ***President*** **Jun 2025 - Present**

A 30-member STEM club using drones, robotics, motors, and computer science to make engineering concepts accessible through hands-on projects and public learning experiences.

- Directed 'DRONEVERSE,' a public drone exhibition reaching 500+ people; partnered with IsLab, a laboratory specializing in drones and AI-enabled drone systems, to deliver expert talks and technical demonstrations.
- Served as a featured student speaker, explaining the ideation, prototyping, and testing journey behind student-led drone and engineering projects.
- Co-organized a CS & STEM Fair at Darahoa charity classroom in Lam Dong Province for 30+ disadvantaged students, introducing computer science, logical thinking, and simple STEM experiments; distributed 30+ gift packages ($390 USD).

PhyX Club / ***Vice President*** **Sep 2025 - Present**

A club making foundational physics engaging for young learners through visual demonstrations, fun experiments, and educational videos.

- Developed the Physics Behind Engineering series, using visual explainers and demonstrations to connect mechanics, energy, electricity, buoyancy, and flight with real engineering applications.
- Organized and taught a two-month educational program at Village Chance for 30 students; delivered hands-on STEM activities and donated $384 USD in school supplies.

Brentwood School / ***Mathematics Mentor*** **Jan 2024 - Present**

- Mentored students in Years 7-10 in algebra, geometry, numerical reasoning, and academic problem-solving through structured peer tutoring.
- Substantially improved a student's grade from a grade 5 to a grade 7 in GCSE Mathematics.

ENGINEERING & INNOVATION

Rescue Drone Engineering Project / ***Developer & Lead Engineer*** **Feb 2025 - Oct 2025**

- Designed, built, and iterated a rescue-drone prototype for emergency supply delivery and search-and-rescue; achieved 20-30 minutes of flight time, a 3-5 kg payload, an approximately 3 km RC range, and extended control through 4G connectivity.
- Completed 60 laboratory test flights and 3 outdoor field trials in Vung Tau and Dong Nai; evaluated stability, endurance, payload performance, and control response to guide design revisions.

Smart Life Jacket / ***Developer & Lead Engineer*** **Mar 2026 - Aug 2026**

- Developed a smart life-jacket concept that reports a wearer's location and estimates drift direction from successive position readings, enabling rescuers to coordinate a search with the rescue drone.
- Specified a candidate stack of a GNSS location module, onboard microcontroller, waterproof enclosure, wireless telemetry, and time-stamped position logs; mapped the jacket-to-drone alert and rescue-team handoff.
- Completed 20 laboratory tests and 2 field trials at Vung Tau Beach and Tri An Lake, Dong Nai, to evaluate the prototype in controlled and open-water settings.
- Set the next development priorities as more reliable alerts, improved waterproofing and battery endurance, and an emergency distress trigger for coordination with rescue teams.

AI, Finance, and the Vietnam Stock Market / ***SVIIF Team Leader*** **May 2025 - Aug 2025**

- Led development of an AI-driven product for analyzing Vietnam equities; processed 600 trading days of VPS API data for VIC, VNM, and VCB and engineered moving-average, price-change, price-range, and lagged-volume features.
- Directed model training and backtesting of Decision Tree trading signals using a 5% stop-loss, historical Value-at-Risk, and portfolio-weight limits adapted to Vietnam's market rules.

ENTREPRENEURSHIP

STEM GEARS Startup / ***Founder*** / ***3D-Printed Cutaway Engine Venture*** **Sep 2025 - Present**

- Founded an educational venture producing 3D-printed cutaway models of aircraft and automobile engines, alongside hands-on STEM kits that help younger students visualize engine components, mechanical motion, and propulsion.
- Led an eight-member team spanning engineering, coding, and design; developed 10 model and kit designs illustrating how different engines operate through visible components and classroom activities.
- Sold 60 products and generated $3,850 USD in revenue; donated the full amount to community initiatives run by Creativism Club and SafeFlow Vietnam.

RESEARCH

Gas Turbine Compressor Optimization Research / ***Co-Researcher, Co-Author*** **Mar 2026 - Oct 2026**

Multi-Objective Optimization Using Deep Learning for Slotted Casing Design in a Small Gas Turbine Engine Compressor
Manuscript submitted to Aerospace (JCR Q2, Engineering, Aerospace).

- Collaborated on a study applying deep learning to multi-objective optimization of slotted casing designs for small gas turbine compressors; contributed to research synthesis, model evaluation, and technical manuscript development.
- The study identified a design increasing stall margin from 14.81% to 25.09%, with an approximately 1.3-percentage-point reduction in design-point efficiency, demonstrating the trade-off between compressor stability and aerodynamic performance.

- Contributed towards editing the manuscript, and developing models of AI to maximize optimization.

BVPAgents: Multimodal AI for Engineering Boundary-Value Problems
Independent Student Researcher **|March 2026 - Oct 2026**
Mentor: Dr. Alireza Ghafarolahi

- Designed and developed BVPAgents using Python and AG2 to investigate whether a multimodal AI system could interpret image-based engineering problems, generate solutions, and verify mathematical and graphical outputs.
- Coordinated five specialized agents for image interpretation, mathematical reasoning, code generation, execution, and verification, integrating GPT-4o and DeepSeek with shared context and iterative feedback.
- Benchmarked the system on 15 boundary-value problems across three difficulty levels, cross-checking outputs against analytical solutions and literature references; obtained successful solutions for 11 problems, including all standard cases.
- Analyzed four unsuccessful cases to identify error propagation, boundary-condition implementation issues, and nonphysical eigenvalues; authored a research manuscript documenting findings, limitations, and proposed improvements to numerical verification.
- Currently being peer assessed by OJSS

EXPERIENCE, ENRICHMENT & OTHERS

Saigon Hi-Tech Park (SHTP) / ***Intern*** **Jun 2025 - Jul 2025**

- Studied how industrial robots, cobots, sensors, and computer vision support automated production; programmed and operated robots during hands-on exercises.
- Practiced electronic-board assembly and troubleshooting and used 3D scanning and printing; learned the basic workflow for semiconductor design, simulation, and testing.
- Contributed to a company-assigned project focused on improving technology used in a garment factory and submitted a report of its findings.

AI and Embedded Systems Training Program / ***HCMUTE*** **Jun 2026 - Sep 2026**

- Completed advanced training in AI and embedded systems applied to autonomous ground vehicles and unmanned aircraft, with an emphasis on sensing, control, and system integration.
- Supported autonomous-vehicle and UAV project work; received favorable feedback from faculty leadership and a recommendation for consideration for direct entry to undergraduate STEM study.

The Duke of Edinburgh's Award / ***Silver-Level Participant*** **Apr 2024 - Aug 2024**

- Completed the volunteering, physical activity, skills-development, and expedition components of the Silver level.

'Under One Sky' Project / ***Core Member*** **June 2024 - Present**

- Volunteered in the project of teaching Vietnamese for ethnic Thai students in Lao Cai province, Vietnam.
- Taught more than 20 students, promoting language learning and civic integration.

SKILLS, LANGUAGES & INTERESTS

- **Programming, Data and Engineering:** Python, Drone prototyping, Flight testing, Systems integration, 3D modeling and printing, Embedded systems
- **Languages:** English, Thai, Vietnamese
- **Sports:** Soccer (4 years captain); Tennis(4 years)